1.3

Systems of DE: 一组 DE

Order: Eq. 中最高阶导

Linear Eq.: (n order:) $a_0(t)y^{(n)} + \cdots + a_n(t)y = 0$

Nonlinear Eq.: (e.g.:) $\theta^{(2)} + \frac{g}{L}\sin\theta = 0$

Linearization: approximating a nonlinear eq. by a linear eq.

Solution: 4 a DE eq. D,
a sol. of it
is a func. ϕ,
whose $\phi^{(0)}, \phi^{(1)}, \cdots, \phi^{(n)}$
satisfy eq. D

(2.1) linear eq. (2.6) exact eq.

(2.2) separable eq.

2.1

1st order linear eq.

form: $y' + p(t)y = g(t)$

$P(t)y' + Q(t)y = G(t)$

Intergrating factor

to determine:

$\mu(t)$: an undetermined func.

$y' + p(t)y = g(t)$

$\mu(t)y' + \mu(t)p(t)y = \mu(t)g(t)$

$\frac{d}{dt}(\mu(t)y(t))$

$\Rightarrow \mu(t)y = \int \mu(t)g(t)\,dt + C$

$\Rightarrow \mu'(t) = \mu(t)p(t)$

$\Rightarrow \frac{\mu'(t)}{\mu(t)} = p(t)$

$\Rightarrow \ln(\mu(t)) = \int p(t)\,dt + C$

$\Rightarrow \mu(t) = e^{\int p(t)\,dt}$

$y = \frac{\int_{t_0}^{t} \mu(s)g(s)\,ds + C}{\mu(t)}$

$\Rightarrow y =$

2.2 (Separable Equations)

(In 1.2: $y' + ay = b$ ← solve — direct integration)

一阶 ODE: $\frac{dy}{dx} = f(y,x)$

↓重写

$$M(x,y) + N(x,y)\frac{dy}{dx} = 0$$

Separable equations:

$$M(x) + N(y)\frac{dy}{dx} = 0$$

$$\int M(x)\,dx + \int N(y)\cdot y'\,dx = 0$$

$$\int M(x)\,dx + \int N(y)\,dy = 0$$

Sol.: Let $m(x)$, $n(y)$ be 2 func.'s

and $\frac{d}{dx}m(x) = M(x)$,

$$\frac{d}{dy}n(y) = N(y)$$

$M(x) + N(y)\frac{dy}{dx} = 0 \Rightarrow \frac{d}{dx}[m(x) + n(y(x))] = 0$

$\Rightarrow m(x) + n(y(x)) = C \Rightarrow m(x) + n(y) - C = 0$

Initial condition: $y(x_0) = y_0$

$\Rightarrow m(x_0) + n(y_0) = C$

$\Rightarrow m(x) - m(x_0) + n(y) - n(y_0) = 0$

$\int_{x_0}^{x} M(s)\,ds + \int_{y_0}^{y} N(t)\,dt = 0$

(solve explicitly 显式求解
solve implicitly 隐式求解)

2、3

(7)

2.4

(Existence & Uniqueness of Sol.'s)

Thm. 2.4.1

{ func. $p(t)$ & $g(t)$ are cont. on an open interval $I:(\alpha,\beta)$

$t_0 \in I=(\alpha,\beta)$

$\Rightarrow \exists y=\phi(t)$, s.t. { $\forall t \in I$, $y'+p(t)y=g(t)$

$y(t_0)=y_0$ 此为 (the initial cond.)

(此处，解 $y(t)$ 可以是不 cont. 的

$\mu(t)=e^{\int_{t_0}^{t} p(s)\,ds}$

$y=\frac{1}{\mu(t)}\left[\left(\int_{t_0}^{t}\mu(s)g(s)\,ds\right)+y_0\right]$

对线性的 ODE

Thm. 2.4.2

{ func. $f(t,y)$ and $\frac{\partial}{\partial y}f(t,y)$ are cont. in a rectangle $\alpha<t<\beta$, $\gamma<y<\delta$

$t_0 \in (\alpha,\beta)$

$y_0 \in (\gamma,\delta)$

$\Rightarrow \exists h$, s.t. $(t_0-h, t_0+h) \subset (\alpha,\beta)$, (①非必需的)

and $\exists y=\phi(t)$ defined on (t_0-h, t_0+h)

①②③

s.t. y is a sol. of the IVP

$\begin{cases} y'=f(t,y) & \text{(the eq.)} \\ y(t_0)=y_0 \end{cases}$

对非线性的 ODE 才

General Sol.

线性方程和非线性方程的另一个不同之处涉及General Solution的概念.

对于一阶线性方程，可以获得包含一个任意常数的解，通过指定该常数的值，可以得出所有可能的解。

对于非线性方程，情况可能并非如此；即使可以找到包含任意常数的解，也可能存在无法通过给该常数赋值而获得的其它解。

因此，general sol. 仅对线性eq.用

Implicit Sol.

找到一个 func. $F(t,y)=0$,

且其解 y 满足 DE 的解 $y=\phi(t)$

2.5

Autonomous

def.: eq.'s of the form:

$y' = f(y)$ (自变量 t 未出现)

$(= \frac{d}{dt} y)$

Exponential Growth

$\frac{dy}{dt} = r \cdot y$ (r is a constant)

r: the rate of growth or decline

$y(0) = y_0$ (initial condition)

$\Rightarrow y = y_0 \cdot e^{rt}$

Logistic Growth

→ replaced by a func. $h(y) = (r - ay)$

Verhulst eq. / Logistic eq.

$$\frac{dy}{dt} = (r - ay)y$$

$\Leftrightarrow$

$$\frac{dy}{dt} = r(1 - \frac{y}{K})y \quad (\text{Eq. } ⑦)$$

r: intrinsic growth rate

the constant sol.'s: $\phi_1(t) = 0$, $\phi_2(t) = K$

↓

equilibrium sol.'s (of Eq. ⑦)

The critical points of eq. $y' = f(y)$

= the roots of eq. $f(y) = 0$ ⑧

2.6 Exact DE

~~Def.~~ (form ⑥) $M(x,y) + N(x,y)\,y' = 0$ (Eq. ⑥)

Sol.'s of Eq. ⑥ are given implicitly by $\psi(x,y) = c$, $\forall$ constant

Thm. 2.6.1

region R: $\alpha < x < \beta$, $\gamma < y < \delta$

Eq. $M(x,y) + N(x,y)\,y' = 0$ is an exact DE

$\Leftrightarrow M_y(x,y) = N_x(x,y)$, $\forall (x,y) \in R$

$\Leftrightarrow \exists$ func. $\psi(x,y)$, s.t. $\psi_x(x,y) = M(x,y)$,

$\psi_y(x,y) = N(x,y)$

Integrating Factors

Cond.: $\mu'(x) = \dfrac{M_y - N_x}{N}\mu(x)$

$\mu'(y) = \dfrac{\dfrac{N_y - M_x}{M}\mu(y)}{0}$

用法: $\mu M + \mu N y' = 0$

$(\mu M)_y = (\mu N)_x$

2.7

Ch. 3 2nd order linear Eq.'s

3.1

A 2nd order ODE

in the form: $y'' = f(t, y, y')$

↑

is linear $\Leftarrow$ $y'' = f(t, y, y')$

$[y'' = g(t) - p(t)y' - q(t)y]$

↓

otherwise: nonlinear

↓

if $g(t) = 0$

$\Rightarrow$ the eq. is homogeneous

↓

otherwise: nonhomogeneous

Characteristic eq.

Def.: For eq. $ay'' + by' + cy = 0$,

it has sol.'s in the form $y = e^{rt}$ ← 待定 coeff.

Then $ar^2e^{rt} + bre^{rt} + ce^{rt} = 0$

$\Rightarrow ar^2 + br + c = 0$

↳ the chara. eq. of the eq. ↑

If the roots of the chara. eq.

are <u>real & different</u>, are r_1 and r_2

---②

⇓

Then $y_1(t) = e^{r_1 t}$ & $y_2(t) = e^{r_2 t}$

are 2 sol.'s of $ay'' + by' + cy = 0$

Now WT: solve IVP $\begin{cases} y(t_0) = y_0 \\ y'(t_0) = y_0' \end{cases}$

↑ the

let $ay'' + by' + cy = 0$

$= (ar_1^2 + br_1 + c)e^{r_1 t}$

$+ (ar_2^2 + br_2 + c)e^{r_2 t}$

$$\Rightarrow \begin{cases} y_0 = c_1 e^{r_1 t_0} + c_2 e^{r_2 t_0} \\ y_0' = c_1 r_1 e^{r_1 t_0} + c_2 r_2 e^{r_2 t_0} \end{cases}$$

$$\Rightarrow \begin{cases} c_1 = \dfrac{y_0' - y_0 r_2}{r_1 - r_2} e^{-r_1 t_0} \\ c_2 = \dfrac{y_0 r_1 - y_0'}{r_1 - r_2} e^{-r_2 t_0} \end{cases}$$

General sol. of $ay'' + by' + c = 0$;

$$y = c_1 e^{r_1 t} + c_2 e^{r_2 t}$$

3.2 (an open interval)

$\forall t \in \underbrace{(\alpha, \beta)}_{=I}$ (可 $\alpha = -\infty$ 及 $\beta = \infty$)

$p(t), q(t)$ are cont. on I

the differential operator L

Def.: $L[\phi] = \phi'' + p(t)\phi + q(t)\phi$

$L = D^2 + pD + q$, where D is the deri. operator

(用 y 表示 $\phi(t)$)

(WT know: If the IVP

$L[y] = y'' + p(t)y' + q(t)y = 0$

$y(t_0) = y_0$, $y'(t_0) = y_0'$

always has a sol.

Thm. 3.2.1 (Existence & Uniqueness Thm.)

(homo. eq., non homo. eq. 都可用)

$$\begin{cases} \text{IVP:} \\ y'' + p(t)y' + q(t)y = g(t) \\ y(t_0) = y_0 \quad , \quad y'(t_0) = y_0' \\ p, q, g \text{ are cont. cont. on an open } \sim I \\ t_0 \in I \end{cases}$$

$\Rightarrow$ $\exists$ exactly only 1 sol. $y = \phi(t)$ of the IVP

(★) ($\Rightarrow$ { sol. exist in this IVP
the IVP has only 1 unique sol.
the sol. $\phi(t)$ of this IVP
is def. on I
& is at lease twice diff. on I))

Eg. 1: find Longest interval ($\sim$)

$\longleftrightarrow$ 找 $p(t)$、$q(t)$ 的 cont. 区间

$\leftrightarrow$ 找 discont. 的点

Thm. 3.2.2 (Principle of Superposition)

$$\begin{cases} DE: L[y] = y'' + p(t)y' + q(t)y = 0 \\ y_1, y_2: \text{2 sol.'s of } \mathcal{I} \end{cases}$$

$\Rightarrow \forall$ constants c_1, c_2,

$(c_1 y_1 + c_2 y_2)$ is a sol. of $\mathcal{I}$

Wronskian (determiant)
of the solutions y_1 & y_2

$W(y_1, y_2)(t_0)$

Thm. 3.2.3

y_1, y_2 are sol.'s of $\underbrace{y'' + p(t)y' + q(t)y = 0}_{\text{Eq. ①}}$

initial condition $\begin{cases} y(t_0) = y_0 \\ y'(t_0) = y_0' \end{cases}$

$W = y_1 y_2' - y_1' y_2 \neq 0$ at t_0

$\Rightarrow \exists$ constants C_1, C_2, $\longleftrightarrow$ 任意 C_1, C_2

s.t. $y = C_1 y_1(t) + C_2 y_2(t)$ $\longleftarrow$ 此为包含 Eq. ① 的所有解

Thm. 3.2.4

$y = C_1 y_1(t) + C_2 y_2(t)$: the general sol. of Eq. ①

y_1, y_2 : a fundamental set of sol.'s of ~

$(W(y_1, y_2)(t_0) \neq 0)$

Thm. 3.2.5

(Eq.①) $y'' + p(t)y' + q(t)y = 0$

$p(t)$, $q(t)$ are cont. on an open interval I

$t_0 \in I$

$y_1(t_0) = 1$, $y_1'(t_0) = 0$

$y_2(t_0) = 0$, $y_2'(t_0) = 1$

$\Rightarrow$ y_1, y_2 form

a fundamental set of sol.'s of ~

($W(\sim, \sim)(\sim) = 1$)

Thm. 3.2.6

(Eq.①) $y'' + p(t)y' + q(t)y = 0$

$p(t)$, $q(t)$ are cont. on an open interval I

$y = u(t) + iv(t)$ is a complex-sol. of ~

$\Rightarrow$ u and v are sol.'s of ~

Thm. 3.2.7 (Abel's Theorem)

$$\begin{cases} y'' + p(t)y' + y = 0 \\ y_1, y_2 \text{ are sol.'s of } \sim \\ p(t),\ q(t) \text{ are cont. on an open interval } I \end{cases}$$

$$\Rightarrow W(y_1, y_2)(t) = c\, e^{-\int p(t)\,dt}$$

$$\begin{cases} c(y_1, y_2) \text{ is a constant depends on } y_1 \& y_2 \\ \forall t \in I, \\ c = 0 \Rightarrow W(y_1, y_2)(t) = 0 \\ c \neq 0 \Rightarrow W(y_1, y_2)(t) \neq 0 \end{cases}$$

3.3

Complex roots of the chara. eq.'s

$ay'' + by' + cy = 0$ (Eq. ①)

$ar^2 + br + c = 0$ (Eq. ②)

↓

if $b^2 - 4ac < 0$,

$\Rightarrow$ roots $\begin{cases} r_1 = \lambda + i\mu \\ r_2 = \lambda - i\mu \end{cases} \in \mathbb{Z}$,

$\lambda, \mu \in \mathbb{R}$

$\Rightarrow \begin{cases} y_1(t) = e^{(\lambda+i\mu)t} \\ y_2(t) = e^{(\lambda-i\mu)t} \end{cases}$

Euler's formula

$e^{it} = \cos t + i \sin t$

$e^{(\lambda+i\mu)t} = e^{\lambda t} \cdot e^{i\mu t}$

$= e^{\lambda t}(\cos \mu t + i \sin \mu t)$

$\Rightarrow$ (of y_1 & y_2:) $\begin{cases} \text{real part: } u(t) = e^{\lambda t}\cos \mu t \\ \text{im. part: } v(t) = e^{\lambda t} \sin \mu t \end{cases}$

Wronskian: $W(u, v)(t) = \mu e^{2\lambda t} \neq 0$

$\Rightarrow$ General sol.: $y = C_1 e^{\lambda t}\cos \mu t + C_2 e^{\lambda t} \sin \mu t$

3.4 Repeated roots; reduction of order

$ay'' + by' + cy = 0$ (Eq. ①)

$ar^2 + br + c = 0$ (Eq. ②)

↓

if $b^2 - 4ac = 0 \Rightarrow$ only root: $r_1 = r_2 = -\frac{b}{2a}$

$\Rightarrow$ 1st sol.: $y_1(t) = e^{-\frac{b}{2a}t}$

WT: find the 2nd sol.

$v''(t) = 0$

$v(t) = c_1 + c_2 t$

$\Rightarrow y = v(t)\, e^{-\frac{b}{2a}t} = (c_1 + c_2 t) \cdot e^{-\frac{b}{2a}t}$

$= c_1 e^{-\frac{b}{2a}t} + c_2 t e^{-\frac{b}{2a}t}$ (General sol.)

$\Rightarrow$ { 1st sol.: $y_1(t) = e^{-\frac{b}{2a}t}$

2nd sol.: $y_2(t) = t \cdot e^{-\frac{b}{2a}t}$

$W(y_1, y_2)(t) = \cdots = e^{-\frac{b}{a}t} \neq 0$

Reduction of order

(参照前页方法，普遍化以找 2nd sol.)

Eq.: $y'' + p(t)y' + q(t)y = 0$

1st. Sol. of ↓: $y_1(t)$

① let $y = v(t)\, y_1(t)$

$\Rightarrow y' = v'(t)\, y_1(t) + v(t)\, y_1'(t)$

$\Rightarrow y'' = v''(t)\, y_1(t) + 2v'(t)\, y_1'(t) + v(t)\, y_1''(t)$

② 代入 $y'' + p(t)y' + q(t)y = 0$

$\Downarrow$

$$\begin{cases} y_1 v'' \\ + (2y_1' + p y_1) v' \\ + (y_1'' + p y_1' + q y_1) v \\ = 0 \end{cases}$$

y_1 是 eq. 的一个 sol.: $y_1'' + p y_1' + q y_1 = 0$

$\Rightarrow y_1 v'' + (2y_1' + p y_1) v' = 0$ (y_1'' 没有了)

3.5

Nonhomogeneous eq.

form (2nd order): $L[y] = y'' + p(t)y' + q(t)y = g(t)$ Eq. ①

$p(t), q(t), g(t)$ are cont. on an open I

homo. Eq. ② ($\downarrow g(t)=0$): $L[y] = y'' + p(t)y' + q(t)y = 0$

Thm. 3.5.1

Y_1, Y_2 are 2 sol.'s of nonhomo. Eq. ①

$\Rightarrow Y_1 - Y_2$ is a sol. of homo. Eq. ②

y_1, y_2 are 2 sol.'s of Eq. ②

(fmd. set ...③)

$\exists$ constants c_1, c_2,

s.t. $Y_1 - Y_2 = c_1 y_1 + c_2 y_2$

Thm. 3.5.2

The general sol. of nonhomo. Eq. ①

can be in the form $y = \phi(t) = c_1 y_1(t) + c_2 y_2(t) + Y(t)$

③ y_1, y_2: 2 sol.'s of Eq. ②

Y: 1 sol. of Eq. ①

寻找 Y(t) (方法×2)

Method of Undetermined Coefficients

(先假设 Y(t) 的形式)

nonhomo. Eq: $ay''+by'+cy=g(t)$

① 求 homo. Eq. 的 general sol.

② $g(t)$ 应为 三角函数/多项式/指数 func.

③ $g(t)=\sum_{i=1}^{n} g_i(t)$

④

⑤

⑥

⑦

Proof: - - -

3.6

(這堂課PPT)

Variation of parameters

nonhomo. Eq: $y'' + p(t)y' + q(t)y = r(t)$

homo. Eq: $\cdots\cdots = 0$

General Theory

Thm. 7.3

I is open, p, q, r are cont. on I

(y_1, y_2) is a set of sol.'s of the homo. Eq.

$\Rightarrow$ Then a sol. of nonhomo. Eq. is:

$$Y(t) = -y_1 \int \frac{y_2 \cdot g}{W(y_1, y_2)}(t)\,dt + y_2 \int \frac{y_1 \cdot g}{W(y_1, y_2)}(t)\,dt$$

general sol.: $y(t) = c_1 y(t) + c_2 y(t) + Y(t)$

Ch.4 Higher order linear eq.'s

4.1

General theory of nth order eq.'s

$$L[y] = y^{(n)} + p_1(t) y^{(n-1)} + \cdots \quad \text{Eq. ②}$$
$$+ p_{n-1}(t) y' + p_n(t) y = g(t)$$

$$\text{IVP: } y(t_0) = y_0, \ \cdots, \ y^{(n-1)}(t_0) = y_0^{(n-1)} \quad ③$$

Thm. 4.1.1

$\begin{cases} p_1, \cdots, p_n, g \text{ are cont. on open } I \\ t_0 \in I \end{cases}$

$\Rightarrow$ Eq. ② with IVP ③ has only 1 sol. $y = \phi(t)$

Homo. Eq.: $g(t) = 0$ ④

Thm. 4.1.2

$\begin{cases} p_1, \cdots, p_n \text{ are cont. on open } I \\ y_1, \cdots, y_n \text{ are sol.'s of homo. Eq. } ⊕ \\ \exists t_k \in I, \text{ s.t. } W(y_1, \cdots, y_n)(t_k) \neq 0 \end{cases}$

$\Rightarrow y = c_1 y_1 + \cdots + c_n y_n$

is the general sol. of homo. Eq. ⊕

fundamental set of sol.'s: ---

$\exists$ constants $k_1, \cdots k_n$,

s.t. $k_1 f_1(t) + \cdots + k_n f_n(t) = 0$

$\Leftrightarrow$ The func.'s are L. dependent

linearly

Thm. 4.1.3

$(y_1, \cdots, y_n)$ is a fund. set of sol.'s of homo. Eq. ⊕ on I

$\Leftrightarrow y_1, \cdots, y_n$ are L. indep. on I

($y_1, \cdots, y_n$ are sol.'s of ...)

pf: ① 这里有证：见课本P247

(by Thm. 4.1.1: sol.为L. indep. 二方有sol.对一IVP, 且该sol.唯一)

Nonhomo. eq.: (Eq. ②) ($g(t) \neq 0$)

Y_1, Y_2 are any 2 sol.'s of nonhomo. eq. ②

$\Rightarrow L[Y_1 - Y_2](t) = L[Y_1](t) - L[Y_2](t)$

$= g(t) - g(t) = 0$

$\Rightarrow (Y_1 - Y_2)$ is a sol. of corre. homo. eq. ⊕

$\Rightarrow$ General sol. of nonhomo. eq. ②:

$y = c_1 y_1(t) + \cdots + c_n y_n(t) + Y(t)$

$\rightarrow$ a particular sol. of nonhomo. eq. ②

4.2 homo. Eq. ①

$L[y] = a_0 y^{(n)} + a_1 y^{(n-1)} + \cdots + a_{n-1} y' + a_n y = 0$

Chara. poly: $Z(r) = a_0 r^n + \cdots + a_{n-1} r + a_n$

Chara. eq.: $Z(r) = 0$

chara. eq. has:

Real & Unique Roots:

(n) sol.'s of homo. Eq. ①: $e^{r_1 t}, \cdots, e^{r_n t}$

distinct (all real, no 2 are equal)

general sol. of homo Eq. ①:

$y = C_1 e^{r_1 t} + \cdots + C_n e^{r_n t}$

Complex Roots:

Repeated Roots:

For an order n eq.,

$Z(r_i) = 0 \leftarrow$ repeated roots

Multiplicity (root appearing times): s ($s \leq n$)

real roots r_i: $e^{r_i t}, t e^{r_i t}, \cdots, t^{s-1} e^{r_i t}$

Complex roots $\lambda \pm i\mu$: $e^{\lambda t}\cos \mu t, \cdots, t^{s-1} e^{\lambda t}\cos \mu t$

$e^{\lambda t}\sin \mu t, \cdots, t^{s-1} e^{\lambda t}\cos \mu t$

4.3

The method of undetermined coefficients

Y: a particular sol.

of nonhomo. eq. $L[y] = a_0 y^{(n)} + \cdots + a_n y = g(t)$

4.4

The method of variation parameters

Y: - - -

L[y] = - - -

$$Y(t) = u_1(t)\,y_1(t) + \cdots + u_n(t)\,y_n(t)$$

determine n func.'s $(u_1, \cdots, u_n)$ ← n conditions

Ch 7 Systems of 1st order linear eq.

7.1 Intro.

Thm. 7.1.1

$F_1, \cdots, F_n$
$\frac{\partial}{\partial x_n} F_1, \cdots, \frac{\partial}{\partial x_n} F_n$ — are cont. on $t\ x_1 \cdots x_n$ space

$(t_0, x_1^0, \cdots, x_n^0) \in R$

$\Rightarrow \exists\ |t - t_0| < h$, s.t. $\exists$!

R: $\alpha < t < \beta$
$\alpha_1 < x_1 < \beta_1$
$\vdots$
$\alpha_n < x_n < \beta_n$

7.7

$\underline{x}' = P(x)\underline{x}$ 的

系统的每个解

基本矩阵 $\Psi(t) = [\underline{x}_1(t) | \cdots | \underline{x}_n(t)]$

diagonalizable

A is dia-able

$\Leftrightarrow \exists T$ s.t. $T^{-1}AT = D$

∴

7.8

7.9

6.1

improper integral $\int_a^\infty f(t)\,dt = \lim_{A\to\infty} \int_a^A f(t)\,dt$

→ exist → conv.

→ x exist → diverge

piecewise cont.

A func. $f(t)$ is pw. cont. on $[\alpha,\beta)$ ($t\in$?)

$\Leftrightarrow \exists n<\infty$ s.t. $\alpha = t_0 < \cdots < t_n = \beta$,

s.t. { f is cont. on $\forall (t_{i-1}, t_i) \subset [\alpha,\beta]$

$\forall i \in [1,n]$, { $\lim_{t\to t_{i-1}^+} f(t) = L < \infty$ (exists and)

$\lim_{t\to t_i^-} f(t) = M < \infty$ (exists and)

Thm. 6.1.1

$f:[a,\infty)\to\mathbb{R}$

$\begin{cases}\forall t\geq a,\ f \text{ is pw. cont. at } t\\ \exists g:[M,\infty)\to[0,\infty),\ M\geq a, \\ \quad \text{s.t.} \begin{cases}\forall t\geq M, |f(t)|\leq g(t)\\ \forall t\geq M, \int_M^\infty g(t)\,dt \text{ conv.}\end{cases}\end{cases} \Rightarrow \int_M^\infty f(t)\,dt \text{ conv.}$

$\begin{cases}\forall t\geq M,\ f(t)\geq g(t)\\ \forall t\geq M,\ \int_M^\infty g(t)\,dt \text{ diverges}\end{cases} \Rightarrow \int_M^\infty f(t)\,dt \text{ div.}$

Laplace transform (of a func. f)

$$\mathcal{L}(f)(s)=F(s)=\int_0^\infty e^{-t\cdot s} f(t)\,dt$$

$$\ker(\mathcal{L}(f)(s))=e^{-t\cdot s}$$

$$\mathcal{L}(c_1 f_1(s)+c_2 f_2(s))$$

$$=c_1\mathcal{L}\{f_1(s)\}+c_2\mathcal{L}\{f_2(s)\}$$

thm. 6.1.2

$f: [0, \infty) \to \mathbb{R}$

$\begin{cases} \forall A > 0,\ f \text{ is pw. cont. on } [0, A] \\ \exists K > 0,\ M \geq 0,\ a \in \mathbb{R},\ \text{s.t. } \forall t \geq M,\ |f(t)| \leq K e^{Mt} \end{cases}$

$\Rightarrow \forall s \in \mathbb{C} \text{ with } \mathrm{Re}(s) > a,$

$\mathcal{L}(f)(s) = F(s) = \int_0^\infty e^{-s \cdot t} f(t)\, dt$ abs. conv.

(split → 2 parts:

$\int_0^\infty e^{-st} f(t)\, dt = \int_0^M \sim + \int_M^\infty \sim$

$\int_0^M \sim$ ↓ finite

$\int_M^\infty \sim$ ↓ depends on growth cond. of $f(t)$

$\forall t \geq M,\ |e^{-st} f(t)| \leq K e^{-st} e^{at} = K e^{(a-s)t}$

e^{-st} ↓ 收斂 ⇐ ↓ 超收斂

($s > a \Rightarrow$ conv.)

6.2

Thm. 6.2.1

If is cont.

$\forall A>0$, f' is pw. cont. on $[0, A]$

$\exists$ constants K, M, a,

s.t. $\forall t \geq M$, $|f(t)| \leq Ke^{at}$

$\Rightarrow \forall s > a$, $\mathcal{L}(f')(t)$ exists

and $\mathcal{L}(f')(t) = s\mathcal{L}(f)(t) - f(0)$

↓

递归使用以找到 $\mathcal{L}(f^{(n)})(t)$

Corollary 6.2.2

$f, f', \cdots, f^{(n-1)}$ are cont.

$\forall A>0$, $f^{(n)}$ is pw. cont. on $[0, A]$

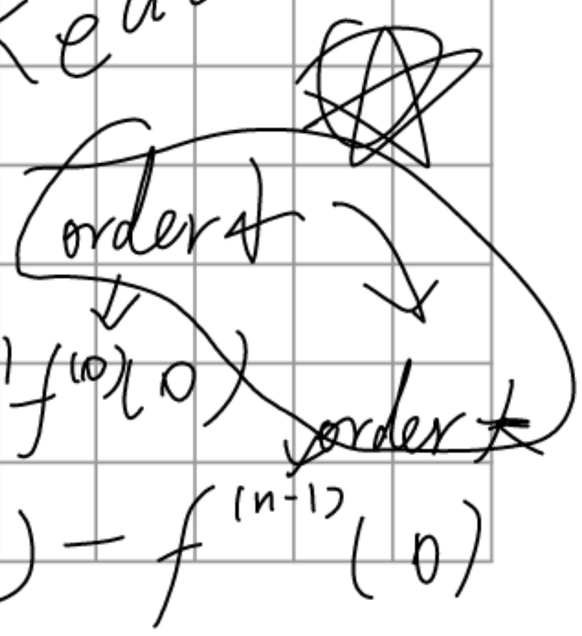
$\exists$ constants K, M, a,

s.t. $\forall i \in [0, n]$, $\forall t \geq M$, $|f^{(i)}(t)| \leq Ke^{at}$

$\Rightarrow \forall s > a$, $\mathcal{L}(f^{(n)})(t)$ exists

and $\mathcal{L}(f^{(n)})(t) = s^n\mathcal{L}(f)(t) - s^{n-1}f^{(0)}(0)$

$- \cdots - sf^{(n-2)}(0) - f^{(n-1)}(0)$



(L. T.)

用于二阶线性ODE：

对于：$ay'' + by' + c = f(t)$　　($y(t)$)

将LT左用于每一个 $y^{(k)}$，以及 $f(t)$

将 $\mathcal{L}(y)(t)$ 表示为 $Y(t)$

$$a[s^2Y(s) - sy(0) - y'(0)]$$

$$+ b[sY(s) - y(0)]$$

$$+ c[Y(s)] = F(s)$$

$$\Rightarrow Y(s) = \frac{(as+b)y(0) + ay'(0) + F(s)}{as^2 + bs + c}$$

↑

($y(s)$ 转为 $Y(s)$，

后可再将 $Y(s)$ 逆转换为 $y(s)$)

6.3

Unit / Step func. / Heaviside func.

$$u_c(t) = \begin{cases} 0, & t < c \\ 1, & t \geq c \end{cases}$$

$$\mathcal{L}\{u_c(t)\} = \int_0^\infty e^{-st} u_c(t)\, dt$$

$$= \int_c^\infty e^{-st}\, dt = \frac{e^{-c\cdot s}}{s}, \quad s > 0$$

Thm. 6.3.1

for func. $f(t)$, $\forall s > a$, $\mathcal{L}\{f(t)\}$ exists

$c > 0$ → ($f(t)$向t軸+平移 c)

$$\Rightarrow \mathcal{L}\{u_c(t)\cdot f(t-c)\}(s)$$

$$= e^{-c\cdot s}\cdot \mathcal{L}\{f(t)\}(s)$$

$$= e^{-cs}\cdot F(s), \quad \forall s > a$$

(Proof: 積分變換 $u = t - c$)

$$\therefore u_c(t)\,\mathcal{L}^{-1}\{F(t-c)(s)\} = \mathcal{L}^{-1}\{e^{-c\cdot s} F(t)(s)\}$$

Thm. 6.3.2

) for func. $f(t)$, $\forall s > a \geq 0$, $\mathcal{L}\{f(t)\}$ exists

$c \in \mathbb{R}$

$\Rightarrow \forall s > a + c$, $\mathcal{L}\{e^{ct} f(t)\} = \mathcal{L}\{f(t)\}(s-c)$

① for $c \in \mathbb{R}$,

$e^{ct} f(t)$ is the inverse Lap. transform of $F(s-c)$

$\Rightarrow \forall s > a + c$ $e^{ct} f(t) = \mathcal{L}^{-1}\{F(s-c)\}$

$\left(\mathcal{L}^{-1}\{F(s)\} = e^{s-t} f(t)\right)$

$t \not{k} = s - c$

$c = s - \not{k} - t$

6.4

求解含有 discont. forcing func.'s 的 DE：

用 Lap. trans.

E.g. 1：

$$2y''+y'+2y=g(t)$$

$$g(t)=u_5(t)-u_{20}(t)=\begin{cases}1, & t\in[5,20)\\ 0, & t\in[0,5)\cup[20,\infty)\end{cases}$$

6.5

Impulse Function

$$ay'' + by' + c = g(t)$$

$d_\tau(t) = g(t)$: $\exists \tau$, 使 $g(t) = \begin{cases} \text{在}(t_0-\tau, t_0+\tau)\text{上很大} \\ 0, \text{otherwise} \end{cases}$

the total impulse of the force $g(t)$ over $(t_0-\tau, t_0+\tau)$:

$$I(\tau) = \int_{t_0-\tau}^{t_0+\tau} g(t)\,dt = \int_{-\infty}^{\infty} g(t)\,dt$$

① $\forall t \neq 0$, $\lim_{\tau \to 0^+} d_\tau(t) = 0$

② $\forall \tau \neq 0$, $I(t) = 1 \Rightarrow \lim_{\tau \to 0^+} I(\tau) = 1$

unit impulse fmction / Dirac delta fmction

$$\begin{cases} \forall t \neq 0,\ \delta(t) = 0 \\ \int_{-\infty}^{\infty} \delta(t)\,dt = 1 \end{cases}$$

$\forall t_0$, a unit imp. func. at t_0

$$\text{is} \begin{cases} \forall t \neq t_0,\ \delta(t-t_0) = 0 \\ \int_{-\infty}^{\infty} \delta(t-t_0)\,dt = 1 \end{cases}$$

Lap. trans:

$$\mathcal{L}\{\delta(t-t_0)\} = \lim_{t\to 0^+} \mathcal{L}\{d_T(t-t_0)\}$$

$$= e^{-s\cdot t_0}$$

$$\mathcal{L}\{\delta(t)\} = \lim_{t_0 \to 0} e^{-s\cdot t_0} = 1$$

$\forall$ cont. fmc. $f(t)$, $\int_{-\infty}^{\infty} \delta(t-t_0) f(t)\,dt$

$$= \widetilde{\quad}$$

$$= f(t_0)$$

6.6

Convolution integral

the convo. of func.'s f and g

def.: $h(t) = (f*g)(t)$

$$= \int_0^t f(\tau)\, g(t-\tau)\, d\tau$$

$$= \int_0^t f(t-\tau)\, g(\tau)\, d\tau$$

(Thm. 6.6.1) $\mathcal{L}\{h(t)\} = F(s)\,G(s) = \mathcal{L}\{f(t)\}\,\mathcal{L}\{g(t)\}$

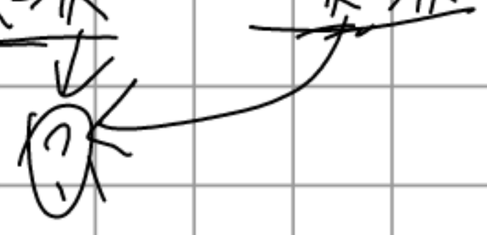
Prop.:

$$f*g = g*f$$

$$f*(g+h) = f*g + f*h$$

$$(f*g)*h = f*(g*h)$$

$$f * 0_{\mathbb{R}\to\mathbb{R}} = 0_{\mathbb{R}\to\mathbb{R}}$$



Transfer function

$$H(s) = \frac{1}{as^2+bs+c}$$

Impulse response $\xrightarrow{=\text{the}}$ inverse Lap. trans. of $H(s)$

$as^2+bs+c=0$

- 不同实根 (r_1, r_2): $h(t) = \frac{e^{r_1 t} - e^{r_2 t}}{a(r_1 - r_2)} u(t)$
- 重复实根 (r): $h(t) = \frac{1}{a} t e^{rt} u(t)$
- 共轭复根 $(\alpha \pm \beta i)$: $h(t) = \frac{1}{a} e^{\alpha t} \sin(\beta t) \cdot u(t)$

解 $\begin{cases} ay'' + by' + cy = g(t) \\ y(0) = y_0 \\ y'(0) = y_0' \end{cases}$

↓

Use Lap. trans.

$$(as^2+bs+c)Y(s) - (as+b)y_0 - a \cdot y_0' = G(s)$$

$$\left(Y(s) = \int_0^\infty e^{-st} y(t)\, dt\right.$$

$$\left. G(s) = \int_0^\infty e^{-st} g(t)\, dt\right)$$

↓

Sol.: $y(t) = \underbrace{\phi(t)}_{\text{homo. eq. sol.}} + \underbrace{\psi(t)}_{= \int_0^t h(t-\tau) g(\tau)\, d\tau}$

9.1

(~ 9.5)

equilibrium sol.:

the sol.'s y s.t. $y' = 0$

critical pt.: 对系统 $\frac{dx}{dt} = Ax$ ---①

critical. pt.'s x_0 s.t. $\frac{dx}{dt} = 0$

trajectory: 当t↑时, x sol. 的轨迹 (phase plane中)

phase plane: 一个 sys. 的 2D 图形表示

phase portrait: 一个 phase plane 中

所有 trajectory 的集合

解 sys. $\frac{d\underline{x}}{dt} = A\underline{x}$ （A: 2×2，$\underline{x}$: 2×1）

对于 A 的特征值，$\det(A-\lambda I)=0$；general sol.

① λ_i 均 { 实 / 不等 / 同号 } ↓

Corre. eigenvector: $\underline{\xi}^{(i)}$

$$\underline{x} = c_1\underline{\xi}^{(1)}e^{\lambda_1 t} + c_2\underline{\xi}^{(2)}e^{\lambda_2 t}$$

$$= e^{\lambda_2 t}\left(c_1'\underline{\xi}^{(1)}e^{(\lambda_1-\lambda_2)t} + c_2'\underline{\xi}^{(2)}\right)$$

node: λ_1 与 λ_2 同号的 $\underline{x}(t)$

- nodal sink: $\lambda_1, \lambda_2 < 0$
- nodal source: $\lambda_1, \lambda_2 > 0$

② λ_i 均 { 实 / 不等 / 异号 }

$$\underline{x} = c_1\underline{\xi}^{(1)}e^{\lambda_1 t} + c_2\underline{\xi}^{(2)}e^{\lambda_2 t}$$

③ λ_i 均 { 实 / 相等 }

eigenvec. indep.

$$\underline{x} = \left(c_1\underline{\xi}^{(1)} + c_2\underline{\xi}^{(2)}\right)e^{\lambda t}$$

eigenvec. dep.

$$\underline{x} = \left[c_1\underline{\xi} + c_2(\underline{\xi}+\underline{\eta})\right]e^{rt}$$

$$\left((A-\lambda I)\underline{\eta} = \underline{\xi}\right)$$

④ λ_i 共为 $\begin{cases} \text{复} \\ u(\lambda_i) \neq 0 \end{cases}$ 设 $\lambda_1 = a+ib$, $\lambda_2 = a-ib$

For system $\underline{x}' = \begin{pmatrix} a & b \\ -b & a \end{pmatrix} \underline{x}$

or $\begin{cases} x_1' = ax_1 + bx_2 \\ x_2' = -bx_1 + ax_2 \end{cases}$

let $\begin{cases} r^2 = x_1^2 + x_2^2 \\ \tan\theta = \dfrac{x_2}{x_1} \end{cases}$

$$\Rightarrow \begin{cases} r'(t) = \lambda r(t) \\ \theta'(t) = -b \end{cases} \Rightarrow \begin{cases} r(t) = Ce^{at} \\ \theta(t) = -bt + \theta_0 \end{cases} \quad (\theta_0 = \theta(0))$$

$$\Rightarrow \begin{cases} x(t) = r\cos\theta = Ce^{at}\cos(-bt+\theta_0) \\ y(t) = r\sin\theta = Ce^{at}\sin(-bt+\theta_0) \end{cases}$$

⑤ λ_i 共为 $\begin{cases} \text{复} \\ u(\lambda_i) = 0 \end{cases}$

For system $\underline{x}' = \begin{pmatrix} 0 & b \\ -b & 0 \end{pmatrix} \underline{x}$

or $\begin{cases} x_1' = bx_2 \\ x_2' = -bx_1 \end{cases}$

$$\Rightarrow \begin{cases} r'(t) = 0 \\ \theta'(t) = -b \end{cases} \Rightarrow \begin{cases} r(t) = C \\ \theta(t) = -bt + \theta_0 \end{cases} \quad (\theta_0 = \theta(0))$$

9.2

Autonomous System

$$\begin{cases} \frac{dx}{dt} = F(x,y) \\ \frac{dy}{dt} = G(x,y) \\ x(t) = x_0 \\ y(t) = y_0 \end{cases}$$

F, G 不显含 t → 自变量与 t 无关

$$\underline{x}' = A\underline{x}$$

$\underline{x} = \underline{0}$

Critical points

for sys. $\underline{x}' = \underline{f}(\underline{x})$, the $\underline{x}_0$ s.t. $\underline{x}_0' = \underline{f}(\underline{x}_0) = \underline{0}$ are [⊥]

critical pt. $\underline{x}_0$ is

stable $\Leftarrow \forall \varepsilon > 0, \exists \delta > 0$,

s.t. $\forall$ sol. $\underline{x}$ where $\|\underline{x}(0) - \underline{x}_0\| < \delta$,

$\forall t > 0, \|\underline{x}(t) - \underline{x}_0\| < \varepsilon$

↓

unstable $\Leftarrow$ ---

asymptotically stable

$\Leftarrow$ { ~ is stable

$\exists \delta_0 > 0$, s.t. $\forall \underline{x}$ where $\|\underline{x}(0) - \underline{x}_0\| < \delta$,

$\lim_{t \to \infty} \|\underline{x}(t)\| = \underline{x}_0$

Suppose pt. $P(x_p, y_p)$

basin of attraction

separatrix

9.3

Thm. 9.3.1

$A \in \mathbb{R}^{2\times 2}$ with eigenvalues λ_1, λ_2

system: $\underline{x}' = A\underline{x}$

$\underline{x}_0 = \underline{0}$ is

$\Rightarrow$ $\underline{x}_0$ is
- asy. stable $\Leftrightarrow$ λ_1, λ_2 都 { 实/复 $u(\lambda_i) < 0$
- stable (不asy.) $\Leftrightarrow$ λ_1, λ_2 都 { 复 $u(\lambda_i) = 0$
- unstable $\Leftrightarrow$ etc. (λ_1, λ_2 都 { 实/复 $u(\lambda_i) > 0$)

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